Neural Bellman Operators
Author Response to the Advisory Referee

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1 Report identity, scientific baseline and present author revision

This is a new author revision of *Neural Bellman Operators*, retaining the original recursive economic decision problem and the Econometrica numerical-methods objective. The completed author baseline is R120 source `39a20bf1fe011175857811bf67bacd788ae80c52`, with its source/PDF publication `cc58ec9613044f97aa1ae5c8d60ec1c3c0021dcd` and final delivery `f06f220aea525088e82228e8e9be4947fb81b1ca`. The R119 and R110 scientific ancestors, their original sources, negative results and incomplete independent episodes remain preserved. A larger revision label is not substituted for a completed source and its evidence.

The fresh 20 October 2026 observation found 361 remote branches and 303 distinct head commits, with no new persisted report or later completed author source. All 360 earlier branch heads are unchanged; the R120 branch has its verified final delivery. The read-only full-tree census enumerated 24,121,986 entries and 67,440 distinct blob identities across those heads. Its compact indexes preserve every commit, tree and stream hash. This is a complete file-identity census, not a claim that every historical program was rerun or every archived proof was freshly revalidated.

The controlling advisory remains report commit `9723c6cd2cc8b3e071422f3e522d4a843946db63`, report blob `33ba8b144e2f7816b19b036404341402d23a582f`, assessing manuscript `f802fe64d91d9f5c1c8f86123b0fbe114e622279`. Its SHA-256 is `479b28c8800084846336cac078877a01dbbd2bae546d179633a2734401da19b9`. It is the same owner-commissioned AI-assisted advisory, not an Econometric Society decision and not a review of R119, R120 or the present revision. All 35 distinct original report bodies, totaling 975,366 bytes, are preserved unchanged under `provenance121/reports121/`. The 36 review references include one reference without a distinct new report. The exact report and author-route custody index is `provenance121/reports121/REPORT_SOURCE_INDEX121.json`.

An author response rather than a change of research program. The literal redirection in M1 is not followed: the author expressly requests continued work on NBO and its original economic objectives at Econometrica. We address its scientific substance by centering the article on one decision problem and one testable mechanism. We do not count a venue change, a different objective, or abandonment of neural operator research as completion. Representation-neutral mathematics is stated as such, and neural-specific economic value remains a question for equal-information and full-cost evidence.

Completed R120 evidence and the unresolved learning mechanism. R120 executed 4,096 primary and 6,144 descriptive timing decisions under its separate source, actual-model and new-state

barriers. Independent original-primitive validation covers all 861 policy keys and 4,096 comparisons, including all-action optimum supports. Its fixed 48-group, 192-event inference family gives 42 positive learned-incumbent event-staircase mean-gain lower bounds, with its stated simultaneous 95% coverage. These are R120 findings.

They also identify the remaining problem. The new paired service differs from inherited refinement at only 12 of 1,024 actions, establishes no strict incremental cell-mean improvement and is slower in all 48 matched clocks. Direct original-model refinement is faster than the paired service in all those clocks. All 512 ReLU–hinge action pairs agree, and every learned completed cell has a strictly worse sample mean original objective than the online control. All 32 three-clock ReLU/full-menu upper-region intersections are empty; 12 individual-clock and two actual-primary upper regions remain nonempty and unresolved. The intersection result does not mean impossibility under every clock. None of these adverse findings is deleted or relabeled as a new R121 result.

New mathematical and algorithmic response. R121 changes acquisition and fitting in the same original economy. The root critic retains the known current action cost and learns the marginal of the integrated original continuation risk in original action units. Theorem 6 derives explicit regularity through binding capacity and constrained adverse-tail cuts. Proposition 7 converts paid original-risk value enclosures into outward marginal labels, including inward boundary probes. Proposition 8 gives exact primal and dual bounds for the actual bounded minimax readout. Theorem 9 transports those labels off design through finite feature ranges, including derivative jumps, and integrates their errors along feasible action paths. Proposition 10 connects an outward stationarity interval to original regret and action distance. The deployed-future allowance is paid once; the recursive value level is not discarded merely because an anchor cancels in a single action comparison.

The development explicitly credits the earlier action-null quotient, convex bounded-readout, constrained-stationarity, active-tail and paired comparison results. The new construction is assessed through its original integrated-risk regularity, actual paid labels, fitted bounded problem, off-design certificate and executed comparison. It is not a claim that quotients, spline or hinge geometry, convex readouts or the basic KKT inequality were invented in this revision.

Prospective numerical scope and current status. The present response is prepared before the new source/protocol freeze and before any original-model R121 outcome. The prospective design has seven stored families and an online control, VALUE and MARGINAL losses within identical family features and bounds, two independently acquired grid budgets and five action nodes with fixed feasible probes. The current catalogue has 58 policy cells, 3,712 primary singleton decisions and 9,280 descriptive decisions in five serial fresh timing processes. The 64 fresh states, rather than risks, methods or clocks, are the statistical units. Every actual first proposal, including a rejected one, and every delivered policy is assigned independent original-primitive evaluation. Fitting is audited through the actual stored coefficients, label endpoints and finite primal/dual witnesses.

Current mathematical sources are `new121/marginal_main.tex` and `new121/marginal_proofs.tex`. The numerical manuscript component `new121/numerical121.tex` and the prospective protocols `PROTOCOL121.md`, `INDEPENDENT_PROTOCOL121.md` and `REPORTING_SPEC121.md` identify the intended study. The completed result text and actual execution receipts will be added only after their own successful execution and read-only verification. No frozen R121 commit, fitted-model artifact, sample, empirical sign, adoption region, independent audit or publication CI success is claimed at this stage.

2 Point-by-point response to the controlling advisory

The following preserves all 30 original requests exactly as quoted in the verified prior author response. Each item distinguishes the completed R120 action, the new R121 action and its continuing scientific boundary. The complete original text, line ranges, report identities, prior author treatment and current routes are retained in `provenance121/R97_REQUIREMENTS121.json`. A planned action and a mathematical subclaim are not blanket closure of the broader request.

2.1 B1. Neural specificity remains unestablished

Original advisory request.

> Validity, recursive qualification, the AVaR enclosure, the information lower bound, and the fallback are representation-neutral. The ReLU family is not the complete-work winner and is never certified adopted over quadratic least squares.

Completed R120 author response. The completed study gives every representation the same four services and original-model information. The ordinary hinge receives the ReLU features and coefficients and pays the full learned-feature acquisition charge. All 512 ReLU–hinge action pairs are exactly equal. All 42 learned own-incumbent cells have strictly positive independent mean-gain lower bounds, but every one of the 14 learned completed cells has a strictly worse sample mean original objective than the online semi-analytic control. The paired service P is slower than inherited L in all 48 matched clocks. Every one of the 32 declared ReLU/full-conventional-menu price intersections across all three clocks is empty, including the intersections of upper not-ruled-out conditions. Thus the achieved contribution is an action-relevant Bellman comparison and original-policy correction; this equal-feature experiment supplies no neural-specific policy or adoption advantage.

The completed evidence for this item is in Table E.I, Table E.II, Table H.VIII; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. R121 changes what the critic learns. Within each stored family, VALUE and MARGINAL receive the same paid original-risk values, including every finite-difference probe, the same features, coefficient box, capacity interval and complete policy service. MARGINAL fits original-action continuation incentives after integrating the risk operator; VALUE fits levels from the identical information bundle. The conventional hinge encoding retains exactly the ReLU features and pays their full charge. Quadratic, spline, polynomial, grid and direct online controls remain in the comparison. The experiment will therefore identify any benefit of the learning objective within a representation before making a cross-representation claim. The finite-box proof is representation-neutral; a neural-exclusive separation has not been established.

Remaining scientific requirement. No neural-specific economic or information advantage is asserted from a common marginal certificate, a fixed feature geometry, or the existence of a neural representation. Such a claim requires an actual equal-information and equal-resource comparison beyond the representation-neutral theorem.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.2 B2. The information-complexity result is too narrow for the platform claim

Original advisory request.

> The theorem is a specialized deterministic one-dimensional oracle result. It does not establish randomized, multivariate, nonuniform-law, or end-to-end Bellman complexity lower bounds.

Completed R120 author response. The new constrained-AVaR theorem supplies the original continuation gradient modulus and the action-distance modulus E. The fixed-known-tail information result has matching width order

$$\min\{KR(1-\alpha)N^{-2},\ ER(1-\alpha)N^{-1}\}.$$

One exact scalar value is one query; the claim is deterministic and univariate. The original smooth model also gives the stronger separate-width coefficient $2M_\alpha R^2(1-\alpha)^2$. The nonsmooth lower-bound adversary therefore does not establish an original-primitives or smooth-class lower bound. Full policy comparisons pay both risk deficits, outward point uncertainty and the unchanged future allowance. In execution the mixed upper is necessary at two recorded accepted levels for one common Poly3 candidate at $m = 4$; no wall-time gain or optimal general Bellman information complexity is inferred from that event. The full positive-margin refinement and future-information-floor propositions state when the interface can terminate successfully.

The completed evidence for this item is in Table H.III; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The new constructive result uses the original known investment primitives to acquire outward integrated-risk values and action-marginal intervals. It is an upper construction for this economy, not a new formula-oracle lower bound. We retain the exact deterministic scalar point-query convention of R119 and the fixed-tail contrast convention of R120, including their function classes, constants and information accounting. The new marginal modulus contains its explicit risk-level and capacity constants. Every label refinement, finite probe and model evaluation belongs to the acquired information bill; supplying the same entire bundle to VALUE and MARGINAL prevents an apparent improvement from unequal access.

Remaining scientific requirement. The generic convex Lipschitz lower bounds do not imply minimax optimality for the smoother closed-form investment class, higher-dimensional risks, or bit complexity. Those extensions require separately stated oracle models and proofs.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.3 B3. The representation result is not decision-theoretic

Original advisory request.

> The ridge example separates restricted approximation classes but does not separate policies, service work, or economic outcomes. A conventional method with the same hinge is equivalent.

Completed R120 author response. The new comparison authorizes changes of the original investment policy. It treats a difference of convex continuations as Lipschitz, without assuming that the difference is convex. The independent original-policy and all-action-optimum witnesses now provide value, gain, regret and action-distance bounds for every returned decision. For ReLU at $\alpha = 1/2$, P's mean original-policy gain over its own completed incumbent is approximately $[212.675, 213.885] \times 10^{-6}$; the mean-regret upper bound falls from 215.906×10^{-6} to 2.626×10^{-6}. The RMS action-error upper bound falls from 0.01470 to 0.001621. L returns the same ReLU policies, and conventional hinge is identical as well. These are original-investment improvements with a shared correction; they neither establish a neural geometry advantage nor borrow economic improvements from the separate posted-price example.

The completed evidence for this item is in Table E.I; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The critic is now measured in the units that affect the original action: the continuation marginal per unit of feasible investment. The finite-cell construction encloses both the true marginal and every feasible one-sided derivative of the fitted feature map. Its integral along an action path controls a Bellman-objective contrast, even for a fitted polynomial or spline without convexity. The capacity-sensitive stationarity bound then connects an outward derivative interval to original regret and action distance. The deployed future-controller allowance is added once. A small node loss is not promoted to a uniform marginal or policy guarantee without the off-design construction.

Remaining scientific requirement. The constructive cell bound may be loose at the executed design. Its usefulness and the resulting original economic decisions must be measured; fitted marginal error alone cannot establish adoption.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.4 B4. The certificate-aware objective fails its intended service connection

Original advisory request.

> It improves residual span in 36 of 36 cells but worsens aggregate rejections and most quadratic cells. The missing theorem is the connection from the trained objective to action-relevant resolution.

Completed R120 author response. The acquired information now targets the original continuation difference at two feasible actions. A paid common-tail mismatch and a strict complete-policy gate connect this information to a permissible decision change, with the true future-controller allowance included once. Holding the three inherited support-gradient proposal rules fixed, P changes 391 of 1,024 actions and accepts 815 steps; L changes 388 and accepts 807. Their final actions differ in only 12 cases, and none of the 16 cell-mean comparisons resolves a strict incremental P advantage. The same paid pair primitive is supplied to the stronger direct conventional proposal. All fitted critics still use the ordinary original-V_1 value-label objective. We therefore complete an action-dependent certificate and its original-policy evaluation, while retaining action-aware training and acquisition as a distinct unresolved scientific requirement.

The completed evidence for this item is in Table E.I, Table H.III; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. R121 implements a different training objective, rather than renaming the residual span. It minimizes the worst discrepancy over outward original-action marginal labels in the same declared finite coefficient box used by the value fit. An exact primal and dual account certifies the stored finite-box fit, with one-sided rows at feature kinks. The subsequent actor remains subject to the original complete-policy service. Independent reconstruction is prescribed for every actual first proposal, including rejected ones, and every final policy. Training discrepancy, acceptance, repair, total work and original regret will be reported jointly, so a rejected learned proposal cannot disappear behind a successful conventional fallback.

Remaining scientific requirement. The finite-box optimum is an optimum of the declared marginal fitting problem. It does not by itself minimize economic regret or complete serving cost, and no R121 empirical improvement is yet available.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.5 B5. Complete work favors a strong conventional method

Original advisory request.

> Quadratic least squares is faster, uses fewer proposal queries, and has no neural acquisition overhead. R97 therefore does not shift the computational frontier toward NBO-specific representation.

Completed R120 author response. The completed four-service/eight-arm comparison retains all 4,096 primary and 6,144 timing decisions and charges actual common-label acquisition, each standalone fit and audit, full hinge acquisition, loading, proposals, tail witnesses, precision ladders, rejected work, fallback and cohort bookkeeping. Summing complete standalone 64-state costs over the same 16 cells gives 94.478 seconds for L, 132.729 for P and 80.302 for direct refinement D. P purchases fewer compiled terminal rows than L, yet is slower in every primary cell and all 48 matched fresh clocks. D purchases substantially more terminal rows and is faster than P in all those clocks. The independent rational study costs a separate 1,806.153 seconds and remains charged as research validation. These complete scopes replace a point-count-only competitiveness claim; they retain the stronger traditional method and every adverse clock.

The completed evidence for this item is in Table E.II; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. Each method pays its standalone acquisition of the complete value-and-probe bundle, feature construction, fitting, optimizer failures, fit verification, model loading, proposal construction, rejected candidates, original-law qualification and fallback. The two state-grid budgets are independently acquired, although their locations are nested. The online control uses the same economic formulas and feasible actions and remains eligible to win the complete-service comparison. Five separate fresh timing processes preserve all observed order reversals. The numerical record will distinguish point-query counts, process time and a complete decision bill; none is used as a substitute for the others.

Remaining scientific requirement. A lower marginal fitting loss or fewer primitive values is not a complete-work victory. R120's stronger direct control and the earlier quadratic winners remain the relevant adverse benchmarks.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.6 B6. The purchase exercise is conditional sensitivity, not economic identification

Original advisory request.

> No learned service is adopted over quadratic least squares. Prices and fees are external, clocks are descriptive, and the policy interval is broad. The exercise does not change an economic conclusion.

Completed R120 author response. The fixed 48-group, 192-event population calculation has actually completed on the prospectively drawn 64 states. Under its stated finite-uniform population and independent-entropy assumption, the simultaneous 95% family gives positive event-staircase mean-gain lower bounds in all 42 learned own-incumbent groups, ranging from 2.661×10^{-6} to 14.192×10^{-6}. The six online groups have zero observed gains and zero lower bounds; the ordinary bounded-mean lower bounds alone are zero. The reported price sets use actual paid work and the declared conventional menu. Positive own-incumbent price regions coexist with empty ReLU/full-menu regions. No calibrated monetary value, preference parameter, market demand or population adoption conclusion is supplied by those normalized-objective calculations.

The completed evidence for this item is in Table E.III, Table H.VIII; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The application continues to report the original recursive objective, independently enclosed policy regret, feasible-action errors, acceptance and full cost. Marginal learning has a direct economic interpretation here: it approximates the incentive to invest under binding capacity and recursive tail risk. Price domains, if reported, will use the same model, sample, error standard and explicit cost convention as the corresponding objective contrast. We preserve the distinction between a sufficient adoption region and an upper region not ruled out by interval uncertainty. Neither the normalized investment example nor a hypothetical price for computer time identifies a calibrated willingness to pay.

Remaining scientific requirement. An economically consequential calibrated application, an externally motivated deployment workload and robust cross-environment adoption still require additional evidence. The current economy cannot establish these by changing its objective or information set.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.7 B7. The new 84 actors lack direct population policy-cost inference

Original advisory request.

R97 transfers only the uniform deterministic qualification bound to the new services. The R96 statistical intervals belong to different frozen actors and cannot establish the new actors' mean-cost ordering.

Completed R120 author response. The source F, actual actors M and single 640-byte state draw D are separately and actually committed before evaluation. The recorded 64 exact 40-bit dyadic states are shared across all arms and risks, and singleton initial-state services prevent a batch fallback from making one row's rule depend on another row. All 4,096 original-policy comparisons and all 48-group/192-event statistics completed under these commitments. The 95% guarantee uses the fixed one-sided binomial allocation and bounded-mean allocation from the frozen protocol; it conditions on the declared independent uniform entropy law and fixed actors. This supplies the new R120 population evidence. It neither completes an old R97 actor family nor imports R98/R115 actors, reserved states or inference into this experiment.

The completed evidence for this item is in Table E.III; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The current design fixes source and reporting rules before acquisition, commits the actual fitted models before drawing the new operating-system entropy, and commits the resulting 64 states before outcomes. Every initial-state service is a singleton, preserving a row-wise policy rule for the declared population. Risks, representations, budgets and repeated clocks share those states and are not additional independent observations. The current inferential family must be fixed in its own reporting specification. The R98, R114/R115, R119 and R120 samples retain their independent identities and are not substituted for this study.

Remaining scientific requirement. A completed finite-catalogue result is not an unconditional population or workload guarantee. The actual new inferential conclusions remain pending, and the broader state-distribution and deployment claims remain separate requirements.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.8 B8. Independent verification stops before fitting

Original advisory request.

Feature selection, interval-aware fitting, rounding, domain validation, and serialization are not independently reproduced. This is the critical new acquisition stack.

Completed R120 author response. The separately implemented rational validator actually evaluates all 861 unique returned policy keys, with 111,069 terminal contexts and 4,096 complete-policy comparisons. Its true Lagrange affine supports also give an original all-action optimum lower witness, rather than attaching an unproved slope to a scalar lower value. Thus the independent calculation covers regret and action-distance bounds as well as policy value. All 4,096 intervals overlap: 726 production intervals contain the independent enclosure, while 3,370 narrower production intervals are contained in it. Only the former direction inherits coverage from the independent enclosure; overlap or the latter direction does not independently prove fine production endpoints. The validator does not import the production terminal compiler, retains its exact rational traces

and 1,806.153-second cost, and claims neither external authorship nor a complete historical fitting-optimizer replication.

The exact completed source/evidence route includes `results120/independent-attempt01/EXECUTION120.json`; the current response map binds all corresponding files.

R121 mathematical and algorithmic action. The numerical plan adds an independent reconstruction of the actual stored readout coefficients, feature rows, paid label endpoints and coefficient bounds. It checks the finite-box primal discrepancy and a feasible rational dual lower bound; a solver status or floating-point KKT residual is not the certificate. At a kink both feasible one-sided rows are retained, and a capacity endpoint uses its inward derivative. The independent original-primitive calculation also covers raw first proposals and delivered policies. These checks extend the audit through the new fitting mechanism instead of stopping after the policy evaluator.

Remaining scientific requirement. This verifies the declared bounded readout problem. It does not certify a global optimum over all neural hidden features, an unbounded approximation class, or an unexecuted higher-dimensional training algorithm.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.9 B9. The active submission remains cumulative and disproportionate

Original advisory request.

> A 47-page article, 37-page supplement, and 230-page appendix are excessive for the narrow current theorem and negative acquisition experiment. Historical preservation need not determine the active submission's length.

Completed R120 author response. The active main argument now follows the original continuation/actor problem, the action-dependent modulus, a usable complete-policy improvement rule, the executed matched comparison and its economic interpretation. The supplement contains every new proof and the independently implemented original-optimum construction. The development tables retain all 64 objective/cost cells, 192 clocks, 192 threshold events, 992 sample contrasts and every prescribed price comparison. The 35 original report bodies and the 1,049 inherited item/parent routes remain accessible with exact R119 source bindings. Their presence does not replace the new evidence: the main text reports P's adverse timing, unresolved incremental economic advantage and the stronger conventional proposal alongside its valid gains and the one observed mixed-bound acceptance.

The completed evidence for this item is in Table E.I; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The main text is organized around one current mechanism: paid original-risk marginal information, certified bounded fitting, feasible action extraction and the complete economic comparison. The complete supplement keeps the original Bellman, gradient, boundary, recursive-utility and information arguments, and the development companion retains historical experiments, failures and report routes. Earlier results explain why learning is changed now; they are not presented as new observations. This organization preserves

the NBO research identity and the Econometrica numerical-methods objective while making the current contribution assessable.

Remaining scientific requirement. Editorial concentration does not close the scientific requirements. The current learning and economic comparison must succeed on its stated evidence, and all unresolved earlier model-specific obligations remain traceable.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.10 B10. The negative result is more compelling than the platform claim

Original advisory request.

> The strongest new fact is that a valid global residual objective can fail to improve delivered decisions. The manuscript should either make that misalignment central or supply a new action-aligned objective and theorem.

Completed R120 author response. The new method buys information about an original action contrast and can change the decision. The continuation modulus, fixed-tail information bounds and quantitative tail-defect/refinement analysis are proved; the completed-incumbent gate and the three support proposals retain their R108–R110 ownership. The completed diagnostics distinguish use from independent contribution: the mixed upper is active at 5,961 of 9,016 paid levels and necessary at two accepted levels representing one Poly3 candidate shared by P and D. The transferred incumbent tail-defect bound is inactive in all 2,290 acquired density pairs. P's final action differs from L in 12 cases, with nine exact cell identities and seven unresolved mean differences; its full cost is higher in all 16 primary cells. We claim the proved comparison and actual fixed-resolution decision example, while preserving the unresolved incremental competitiveness and the unfinished action-aware acquisition problem.

The completed evidence for this item is in Table H.III, Table E.II; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The adverse findings drive a substantive algorithmic change. R97's span-oriented fits improved an internal certificate while increasing rejected first actions. R120 obtained certified own-incumbent gains, yet the new paid pair service was slower than the inherited service in all matched clocks and did not establish a strict incremental mean gain. R121 therefore acquires and fits the original continuation marginal, rather than retiming those services or excluding their winners. The full new first-proposal and final-policy census will retain adverse actions, failed fits, rejections and conventional victories.

Remaining scientific requirement. The motivation is not a completed positive finding. If marginal fitting fails to improve the original economic and computational frontier, that failure remains a result and a concrete constraint on the next algorithm.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.11 M1. Recast the paper around one method-neutral question

Original advisory request.

> Ask when a stored continuation can be safely purchased for recursive-risk dynamic decisions. Treat neural networks as one optional representation, not the source of validity.

Completed R120 author response. Neural Bellman Operators remains the same original-economic-decision author project. The main question is how learned continuations, recursive feasible actors and action-relevant information can produce valid improvements at justified complete cost. The investment model now supplies actual policy changes, original-value/regret bounds and fixed-law gain inference, rather than an internal fit metric alone. Its continuous shock changes stock imbalance, while investment trades current adjustment cost against the original risk-sensitive continuation and capacity constraint. The main interpretation also confronts the strong online and direct controls and full paid work. The new safeguard is shared across representations; those facts preserve the neural Bellman research identity without assigning an unsupported architecture-exclusive mechanism.

The completed evidence for this item is in Table E.I; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. We address the scientific request for a single question: which continuation information should an NBO learn to improve a constrained decision under the original recursive objective and a paid numerical budget? We do not follow the literal recommendation to abandon the NBO framing or redirect the paper to a different subject or venue, because the author's instruction expressly preserves that research program and the Econometrica objective. The main text instead gives the requested concentration through an original-risk marginal critic, finite-box training certificate and complete economic test. A representation-neutral theorem is distinguished from any neural-specific empirical contribution.

Remaining scientific requirement. The original ambition remains. Neither a venue change, a new economic objective, nor a decision to stop seeking neural computational value is counted as a scientific response.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.12 M2. Establish the true novelty of the information theorem

Original advisory request.

> Give a theorem-by-theorem comparison with information-based complexity. Add a genuinely new randomized, nonuniform-law, multivariate, or Bellman-compositional result, or present the current theorem as a specialized supporting lemma.

Completed R120 author response. The novelty statement distinguishes the new original constrained-AVaR smoothness calculation, action-distance integral interface, matching fixed-tail query order and quantitative constrained-tail transfer from the inherited common-risk identity and strict policy gate. The original model's additional smoothness gives its own stronger upper coefficient and is

excluded from the generic nonsmooth lower-class claim. The full finite-resolution theorem includes paid deficits and the true future-policy floor. Actual execution shows one Poly3 acceptance at a paid resolution that the separate convex upper cannot authorize at that same resolution. It does not measure a traditional method after another refinement, a counterfactual time saving, or general end-to-end Bellman optimality. The economic and timing comparisons retain those precise limits.

The completed evidence for this item is in Table H.III; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. We state the lineage of the mathematical ideas explicitly. R52 already developed action-null quotient and incumbent-acceptance arguments; R67 already proposed convex certificate-aware bounded readout with a dual account; R99 supplies the constrained quadratic stationarity inequality. R119 proves the tail-uniform exact-query order, and R120 proves the paid action-scaled comparison and tail transfer. The new work derives the original integrated-AVaR marginal through constrained tail cuts, constructs paid outward marginal labels, certifies the actually fitted bounded readout, and transports it off design through feature ranges that retain derivative jumps. Its novelty is assessed at that precise combination and implementation.

Remaining scientific requirement. No generic lower bound for known economic formulas, no first-ever quotient or convex readout claim, and no neural-exclusive information theorem is asserted. Broader oracle and recursive-model results remain separate mathematical tasks.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.13 M3. Develop an action-aligned acquisition objective

Original advisory request.

> Construct a loss that controls rejection, endpoint work, or policy-cost resolution on a predeclared context law. Prove the connection and validate it prospectively against complete work.

Completed R120 author response. All 16 stored model/risk artifacts were freshly acquired under the frozen ordinary original-V_1 interval-midpoint value-label rule, with 256 common sites per risk and complete fitting/audit costs. The new action-relevant acquisition occurs inside the paid policy-comparison interface; it does not change that training objective. Its original-policy consequences are independently evaluated, including 42 positive learned-incumbent gain groups and the unresolved P-versus-L incremental mean comparison. The item therefore separates the completed comparison-information theorem from the still-needed action-aligned loss, acquisition theorem and newly fitted-actor validation. R98's distinct seven-objective acquisition experiment remains preserved under its own unchanged actor and sample barriers.

The completed evidence for this item is in Table E.I, Table E.II; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. Acquisition is now directly connected to the intended action marginal. At five preassigned normalized action nodes, the program buys original-risk

value intervals at the node and its fixed feasible probes. Forward, backward and centered strong-convexity inequalities yield outward derivative labels; boundary nodes use inward probes. The finite-box minimax loss uses those labels, and the same paid values and probes remain available to the value fit. The finite-cell and path-integral construction supplies an explicit route from the acquired information to a policy-loss account. No probe position, grid size or coefficient box may be selected retrospectively from test outcomes.

Remaining scientific requirement. The prescribed design changes acquisition and fitting, but it is not a proved optimal adaptive acquisition policy. The achieved off-design error, paid cost and original decision gains remain empirical questions.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.14 M4. Expand the conventional geometry frontier

Original advisory request.

> Compare ReLU with conventional oblique hinges, adaptive triangulation, convex regression, and representation-neutral basis selection under equal information, rank, acquisition, and validation budgets.

Completed R120 author response. The contemporary frontier contains online semi-analytic, quadratic, ReLU, equal-feature hinge, spline, degree-two polynomial, degree-three polynomial and grid arms, each with C, L, P and D. All 992 unordered within-risk cell comparisons are retained. Every learned completed cell is strictly worse than online in sample original value, and all 512 ReLU–hinge policy pairs are identical under full equal acquisition charges. D has a strictly better independent mean than L and P in five cells, two exact online identities and nine unresolved comparisons; it is faster than P in every matched clock. Thus conventional geometry, direct original-formula access and the shared certificate all receive their actual competitors. A favorable comparison with only a neural arm's own incumbent is not used as the representation frontier.

The completed evidence for this item is in Table E.I, Table H.VIII; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The complete comparison retains seven stored representation families and an online original-model control. Within each family, VALUE and MARGINAL have identical features, bounds and paid data. The ReLU and conventional hinge encodings share the same actual feature geometry and coefficients where prescribed, and both pay the full feature charge. Quadratic, spline, degree-two and degree-three polynomial, and multilinear-grid alternatives remain explicit. Different feature counts, coefficient restrictions and fit capacities are reported rather than described as identical approximation classes. The two nested grid budgets are separately acquired; no larger-budget information is supplied free to a smaller-budget arm.

Remaining scientific requirement. A gain available to a hinge, spline or polynomial is not an architecture-exclusive neural gain. Actual complete-work and economic superiority must be judged against the entire feasible conventional menu.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.15 M5. Treat the negative result as first-class evidence

Original advisory request.

> Report aggregate and cell-level rejection changes beside span improvements, explain where the global objective allocates approximation effort, and do not label 36/36 span reduction as delivered-service success.

Completed R120 author response. The main text and complete development tables retain P's 16 primary cost losses and all 48 fresh-clock losses to L, nine exact and seven unresolved P/L mean comparisons, and all six zero-gain online population groups. P retains 2,257 rejected candidates, including 785 exhausted precision ladders; D retains 633 rejections, including 194 exhausted ladders. The inactive transferred-deficit result and the distinction between 5,961 active mixed bounds and only one necessary economic candidate are reported directly. The older R97 residual-span mismatch, adverse R119 phases, conventional wins and unverified neural-adoption regions remain unchanged. No discarded clock, unresolved sign or failed attempt is removed to make the new mechanism appear uniformly favorable.

The completed evidence for this item is in Table H.III, Table E.II; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The development record preserves the R49 original-policy losses, the later exact quadratic and hinge benchmarks, R97's rejection increase, every adverse A/B/C phase in R119, and the complete R120 pair-service and adoption findings. The new census explicitly records each actual first proposal before qualification and includes its independent original-objective evaluation even when it is rejected. Thus a good final conventional repair does not conceal an economically poor learned proposal. Failed acquisition, fitting, proof diagnostics, execution and publication attempts retain their own actual sources and scopes.

Remaining scientific requirement. No old negative is erased, recoded as a success or replaced by a fresh sample. The current study may sharpen an unresolved failure mechanism without closing the overall computational contribution.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.16 M6. Execute a consequential application or narrow the claim

Original advisory request.

> Either calibrate policy margins, compute prices, fees, and workloads in an application where adoption matters, or state that the paper is about numerical certification rather than economic purchase.

Completed R120 author response. The economic interpretation reports the original risk-sensitive investment objective, not a different loss or welfare surrogate. It shows how the independently bounded ReLU own-incumbent gains reduce original regret and action-distance ceilings, and why a larger spline gain can reflect a worse incumbent. The online semi-analytic policy is already strictly better than every learned completed policy on the sample, while direct refinement is a material traditional competitor. The simultaneous 42 positive learned-incumbent lower bounds and all actual price scenarios are reported with their population and cost scopes. The units remain normalized original-objective units: this two-date experiment adds decision-relevant evidence but does not claim monetary, consumption-equivalent or field-adoption calibration.

The completed evidence for this item is in Table E.I, Table E.III; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The economic mechanism is stated in original investment units. The learned object is the risk-adjusted marginal continuation cost; together with the known current action cost it determines whether the optimal action is interior, zero or at capacity. The theory includes moving adverse tails and active boundaries. Independent original-primitive evaluation compares both raw incentives expressed as proposals and the delivered policy objective. This makes a numerical improvement interpretable as an investment decision and its recursive cost, rather than a normalized fitting score.

Remaining scientific requirement. The finite normalized investment study remains distinct from the unreproduced NDU, continuous-game, Epstein–Zin and high-dimensional application claims. A calibrated, substantively consequential external application is still an open requirement.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.17 M7. Add a genuine scaling experiment

Original advisory request.

> Vary horizon, controls, innovation dimension, and risk level under matched comparators. Report product-point counts, memory, endpoint work, acquisition, and complete service work.

Completed R120 author response. The completed experiment is explicitly two-dimensional, two-date and evaluated at the two prescribed risk levels. It reports each service's full acquisition, original-value queries, candidate/refinement counters, raw process costs and complete resource-vector frontier. The four predeclared amortization volumes are conditions computed from these retained measurements; they are not measured throughput, higher-dimensional scaling or a horizon experiment. The new information theorem has a scalar-tail scope, and the generic lower bound excludes richer original-formula access. Existing higher-dimensional and application evidence retains its own source and limitations. Broader scaling and consequential application claims remain concrete research obligations rather than being filled by these 64 new initial states.

The completed evidence for this item is in Table E.II, Table H.VIII; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The current experiment varies a concrete paid acquisition scale: the 5^2 and 9^2 state grids, with five action nodes and their probes, require 325 and 1,053 original-value contexts per risk before precision refinements. Both budgets are separately acquired and all families receive their entire respective bundle. The theoretical cell construction shows what an off-design guarantee costs in feature ranges and target moduli. These two finite resolutions will be reported as an information-budget comparison in the original two-state, two-date model, with actual work and memory where observed.

Remaining scientific requirement. Two resolutions do not establish dimension, horizon, tensor-rank or risk-dimension scaling. Old high-dimensional and longer-horizon episodes retain their own code, evidence and unresolved requirements; this study does not replace them.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.18 M8. Strengthen timing evidence

Original advisory request.

> Use more repetitions, record or control processor state, and make operation counts primary. Predeclare the inferential target for deployment clocks if a probabilistic purchase claim is desired.

Completed R120 author response. All three 32-state timing processes completed, retaining 6,144 timed actions and 192 cohort clocks. Every timed action, interval and recorded query count matches its corresponding primary rule; no mismatch or clock was censored. P is slower than L in all 48 matched clocks; D is faster than P in all 48 and faster than L in 41, with seven losses and two adverse cell medians. Cohort measurements, standalone acquisition charges, loading and larger process/launcher scopes are labeled separately. These are descriptive engineering observations under the observed execution conditions. The clocks are not extra population observations and do not supply confidence bounds for future hardware, latency, energy prices or external load.

The completed evidence for this item is in Table E.II; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. Five serial fresh timing processes apply the same declared service catalogue to the fixed timing states. The record includes cold process and model loading, standalone acquisition, all fitting and verification, rejected proposals, refinement and complete returned decisions. Timing states and repeated clocks do not enlarge the population sample. The full set of clocks and order reversals remains visible; a favorable median is not a confidence interval or a cross-hardware performance theorem. Primitive-query counts are reported as their stated interface counts, not as unmeasured floating-point operations.

Remaining scientific requirement. A calibrated deployment workload, process variability across machines and universal amortization rates are not identified by these five clocks. Their cost evidence is conditional on the actual recorded environment and complete-service convention.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.19 M9. Independently reimplement fitting and package a standalone environment

Original advisory request.

> Reproduce feature selection, interval-aware fitting, rounding, whole-domain validation, and serialization independently. Include a container or lockfile and all inherited modules needed to run the new tests from the artifact alone.

Completed R120 author response. The exact source F, actual actor M, new sample D and pre-outcome reporting specification R are preserved together with their real launch and publication receipts. Primary evidence was separately published as E0 before final reporting. Independent original-primitives validation, the fixed rational statistics and the first read-only reporter have all completed with exit zero; the reporter binds 777 unchanged input files and 36 stable products. The derivative algebra diagnostic preserves its first source-path failure and the corrected second attempt, without changing frozen science. The bounded package audit and source CI distinguish record/byte checks and statistical/report replays from a fresh primitive-economy rerun. Actual final source, PDFs, build logs and CI states belong to the separate source/PDF and publication declarations; no R119 success or future R120 job is asserted here as a completed R120 CI.

The exact completed source/evidence route includes `provenance/SOURCE_PUBLICATION120.json`; the current response map binds all corresponding files.

R121 mathematical and algorithmic action. The independent fitting audit reconstructs the bounded primal problem and an exact feasible dual witness from stored coefficients, information endpoints and actual feature rows. The independent economic evaluator reconstructs the original primitives for all distinct final and first-proposal keys, then reaggregates every declared comparison. A standalone package will bind the frozen sources, actual data/model barriers, all raw attempts, report outputs and build receipts. Archive byte checks, record invariants, bounded reporting replay and an independent primitive reconstruction have distinct scopes; none is labeled as an entire scientific rerun unless it actually runs.

Remaining scientific requirement. The finite readout audit is not a global hidden-feature optimization certificate, and publication or CI success cannot be claimed before its source-bound receipt exists.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.20 M10. Shorten the active submission substantially

Original advisory request.

> Move historical material to an archival companion. The active article should contain only the current qualification theorem, precise information result, acquisition objective, matched experiment, and supported economic interpretation.

Completed R120 author response. The main paper concentrates on the original recursive decision, complete new mathematical argument, contemporaneous experiment and economic interpretation. Full assumptions, proofs, independent optimum verification and all detailed tables remain in the supplement/development companion. All 35 exact original reports are copied unchanged into the current source package; the 620 major, 394 additional and 35 overlapping-parent treatments retain their original identities and specific remaining requirements. The current response supplies an actual proof/evidence route for each of the 30 controlling items. This organization preserves NBO's learned continuation and actor problem, the strong traditional comparisons and every valid historical result, while keeping the current contribution assessable without treating document reorganization as scientific closure.

The completed evidence for this item is in Table E.I; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The English main text presents the original economic problem, new original-risk marginal construction, certified fitting and action extraction, then the complete current comparison and its economic interpretation. Full proofs remain in the supplement. Dated prior contributions, every original report and their exact response routes remain in the development and provenance records. The 35 distinct reports retain 1,049 authored route records, including overlapping parent treatments; this is not a claim of 1,049 independent issues or of universal closure. The current response quotes each of the 30 controlling requests verbatim and records its remaining scope.

Remaining scientific requirement. Moving development history out of the main argument changes presentation only. It does not delete a theorem, erase a negative result, or resolve a report item by summary.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.21 Minor 1. Error-versus-information terminology

Original advisory request.

> 1. State 'N^{-2}' consistently as an error-versus-information rate; "information exponent two" is easy to misread.

Completed R120 author response. The new result is called interval width versus scalar point information, with order $\min\{KR(1-\alpha)N^{-2}, ER(1-\alpha)N^{-1}\}$. Original-policy contrast, certified gain, regret, value width, squared action-distance bound and complete work are separate quantities throughout the proofs and all 64 objective/cost cells. The independently computed optimum is an enclosure, so the displayed RMS quantities are upper bounds rather than measured errors to a known exact optimizer. The achieved R119 N^{-2} tail-uniform AVaR width result retains its established attribution; the executed economic gains require their own policy evidence.

The completed evidence for this item is in Table E.I; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. We separate the quantities throughout: point-query information, AVaR interval width, original-action marginal error, finite-box fitting gap, integrated action-contrast error, policy regret and action distance. The retained N^{-2} width theorem has its exact oracle class and risk-level constants. The new finite-probe and feature-range bounds are translated into regret only through their displayed path or stationarity inequalities. No fitted-loss or width order is relabeled as an observed economic rate.

Remaining scientific requirement. Any empirical rate estimate requires its own executed resolution series and declared fitting/evaluation standard. The two planned information budgets do not by themselves establish an asymptotic rate.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.22 Minor 2. Oracle qualifiers

Original advisory request.

> 2. Put "deterministic, point-value, univariate, fixed-alpha" adjacent to every high-level summary of the lower bound.

Completed R120 author response. Deterministic, exact scalar point-value, univariate, uniform-law, convex-Lipschitz and fixed-known-tail qualifiers appear with the new information statement. Purchasing the two continuation values at a point costs two queries. The original smooth investment formula is a narrower class than the lower adversary, and risk-tail localization, finite point precision, terminal solves and the future controller remain separately paid. R119's risk-level-uniform width theorem is retained under its own assumptions. The completed two-risk experiment does not establish randomized, multivariate or arbitrary-law oracle optimality.

The completed evidence for this item is in Table H.III; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The inherited lower bounds retain the deterministic, exact scalar-point, univariate query convention, their function classes and the stated range of the query budget. Fixed-tail bounds retain their separate tail-selection and mismatch costs. The current learning experiment uses known original economic formulas with paid outward evaluations and gives the identical acquired bundle to both losses. It makes no oracle-lower-bound claim for formula access, vector batching, hidden analytic structure or randomized information.

Remaining scientific requirement. The current exact formula and smoothness structure can permit stronger conventional algorithms than the generic point-query model. Those algorithms remain eligible controls, and the generic lower bound cannot rule them out.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.23 Minor 3. Tail dependence

Original advisory request.

> 3. Discuss the opposite tail-level behavior of the lower and upper constants as 'alpha' approaches one.

Completed R120 author response. The achieved R119 order $KR(1-\alpha)N^{-2}$ is preserved. R120's fixed-tail comparison intersects widths bounded by $2KR(1-\alpha)/n^2$ and $ER(1-\alpha)/n$, with the original smooth coefficient supplied separately. The action modulus includes its explicit $\alpha/(1-\alpha)$ dependence; this prevents a small tail mass alone from implying a cheap original-model computation near $\alpha = 1$. The full-support $\alpha = 0$ boundary is separate and the normalized-tail statement excludes $\alpha = 1$. Actual new observations use only the predeclared $1/2$ and $3/4$ risk levels and retain every paid tail/terminal operation.

The completed evidence for this item is in Table E.II, Table E.I; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The original marginal theorem displays the dependence on the AVaR tail mass, the strict shock curvature and the investment-capacity range. Constrained tail cuts are included explicitly rather than replaced by an interior-only derivative formula. The finite-difference width includes the paid value precision divided by probe distance. We retain the achieved R119 tail-uniform value-width result separately: its cancellation does not automatically make the new marginal modulus, coefficient fitting work or bit complexity uniform as the risk level approaches one.

Remaining scientific requirement. The experiment uses the two fixed risk levels of the original benchmark. Extreme-risk derivative complexity and a tail-uniform marginal-learning algorithm remain separate questions.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.24 Minor 4. Acquisition restrictions

Original advisory request.

> 4. The readout coefficient bounds are acquisition restrictions, not characterizations of the best approximation; report sensitivity.

Completed R120 author response. The actual actor artifacts and acquisition table preserve all prescribed feature counts, coefficient signs/bounds, rounding and ties. The constrained rank-six readouts are compared with the larger fixed polynomial and grid controls under the same 256 original-V_1 label sites per risk. ReLU and ordinary hinge have identical retained parameters and actions, with the full shared-fit charge assigned to each standalone deployment. These are acquired-model comparisons under known restrictions. They are not best-approximation results for unrestricted representation classes, and the experiment supplies no post-outcome coefficient or architecture sensitivity menu.

The completed evidence for this item is in Table E.I, Table E.II; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. Every finite coefficient box and sign restriction belongs to the approximation problem and is disclosed. VALUE and MARGINAL use the same box and feature map within each family. The primal/dual result certifies that declared bounded problem, not the best approximation over all weights or over an unbounded architecture. Both acquisition budgets and the kink-side conventions are fixed before outcomes; neither the box nor the feature family is enlarged afterward to rescue an unfavorable fit.

Remaining scientific requirement. A finite-box fit can be accurately solved while the representation bias remains large. Its actual original policy and complete cost, rather than the solver gap alone, determine economic usefulness.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.25 Minor 5. Residual span and action gap

Original advisory request.

> 5. Distinguish validated residual span from Bellman action gap in terminology and tables.

Completed R120 author response. Every primary group now has separate original-policy bounds, independent original-optimum bounds, mean regret, RMS action-error upper bound and own-incumbent gain, alongside production contrast and certificate counters. The independent validator supplies 55,104 retained all-action support rows and all 4,096 policy comparisons. For example, the ReLU P gain is economically positive against its incumbent while its mean difference from L is exactly zero. A narrower value interval, smaller fit residual or lower regret ceiling is therefore never counted as an observed policy-value decrease by itself. The 726 versus 3,370 containment directions also retain their different independent-verification implications.

The completed evidence for this item is in Table E.I; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The main theorem now measures action contrasts by integrating a certified marginal error over the actual feasible action path. A state-only anchor cancels within that action comparison, but is not discarded from a recursive value computation. Boundary-sensitive stationarity gives a further original regret and action-distance bound when its conditions hold. The numerical record keeps residual or value discrepancies separate from these quantities and independently evaluates the proposals and delivered policies.

Remaining scientific requirement. A better span, a better node fit or a smaller primal gap need not rank the same policies as the original economic objective. That distinction remains explicit if the new experiment is adverse.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.26 Minor 6. Rejections and unsuccessful work

Original advisory request.

> 6. Report rejection changes beside every span improvement.

Completed R120 author response. Every attempted candidate and resolution, rejection, exhausted ladder, unchanged action and fallback is retained in the actual traces and complete tables. P's 2,257 refusals and D's 633 refusals appear beside accepted-step counts, as do 785 and 194 exhausted ladders. Mixed-bound use is reported at all 9,016 paid levels: 5,961 active upper bounds and two necessary accepted records for one economic candidate. The transferred incumbent bound is inactive in all 2,290 density pairs. Its mathematical validity is not converted into an unobserved reduction of already purchased tail endpoints, candidate-upper work or total time. Every unresolved and adverse comparison remains included.

The completed evidence for this item is in Table H.III; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The new evidence census starts before the acceptance gate. It retains the fitted-loss and primal/dual diagnostics, the actual first proposed action, its independent original-policy value, the acceptance decision, any rejected work, repair and the delivered policy. Reporting both the first proposal and the complete service directly tests whether a fitting improvement aligns with the economic decision. A rejected learned proposal is still part of the method's scientific and cost record.

Remaining scientific requirement. The number of fitting improvements, accepted candidates or qualified final outputs is not itself a welfare comparison. No candidate is censored because its independently evaluated regret is large.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.27 Minor 7. Observed engineering ranges

Original advisory request.

> 7. Label the acquisition bracket as an observed engineering range, not a confidence interval.

Completed R120 author response. The three process-specific measurements, all 192 cohort clocks and their medians and ranges are labeled descriptive engineering observations. All 6,144 comparisons of actions, intervals and query counts match, so the reported conditional pairing of timing and economic outcomes is actually checked. These clocks remain distinct from outward rational value enclosures and from the explicitly proved simultaneous 95% population family of 48 groups and 192 events. Timing replicates do not increase its 64 independent initial-state units. The inherited seven R119 clocks retain their separate episode and do not enlarge the current sample.

The completed evidence for this item is in Table E.II, Table E.III; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The five fresh clocks are retained as measurements of engineering variability under the declared workload. Their extrema and medians are not called sampling confidence intervals. Statistical coverage for the independently sampled states uses only the prospectively fixed inferential family and its assumptions. Complete cost includes probe acquisition, fitting, verification, loading and all accepted and rejected service work. Clock-specific and combined price statements retain their exact intersection conventions.

Remaining scientific requirement. There is no universal timing or economic-price distribution in the current design. Deployment conclusions remain conditional on the recorded workload and environment.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.28 Minor 8. Adoption over the conventional frontier

Original advisory request.

> 8. State visibly that there are zero adoption regions over quadratic least squares.

Completed R120 author response. The old zero-adoption-over-quadratic result and the R119 conventional frontier remain explicit. In the new fixed comparison, positive gains against learned incumbents coexist with an online advantage against all 14 learned completed means and exact ReLU–hinge identity. All 32 ReLU intersections against the predeclared full conventional menu, across four amortization volumes and three clocks, have empty sufficient-price sets and empty intersections of upper not-ruled-out sets. Among individual clock comparisons for ReLU, 12 of 96 upper regions remain nonempty; two of eight actual-primary workload upper regions are also nonempty. These are unresolved comparisons, retained separately from the three-clock intersections. The 17 nonempty sufficient menu regions across representations belong to completed conventional policies. The 112 separate actual-64-state price domains retain the actual workload and charges. These strict recorded-menu conditions and own-incumbent gains do not assert population adoption or rule on every possible deployment criterion.

The completed evidence for this item is in Table H.VIII, Table E.I; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The R97 zero certified adoption findings remain historical facts at their own design. The completed R120 evidence also remains exact: all 32 three-clock ReLU/full-menu upper-region intersections are empty, while 12 individual-clock upper regions and two actual-primary upper regions are nonempty and unresolved. We do not turn the intersection statement into impossibility under every separate clock. R121 changes acquisition and fitting under a new frozen protocol; its original-policy and full-cost regions must be computed from its actual new data without prespecifying a favorable sign.

Remaining scientific requirement. Neither an unresolved upper region nor an own-incumbent gain certifies adoption over the full conventional menu. Current R121 regions are pending.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.29 Minor 9. Verified approximation and HJB literature

Original advisory request.

> 9. Compare the certification contribution carefully with existing verified approximation and HJB-residual literatures.

Completed R120 author response. The main text locates the mixed integration interface relative to retained verified approximation, Bellman/HJB residual ideas, constrained safeguards and the source-owned common-risk identity. Complete proofs retain the original filtration, boundaries and one-sided future-policy allowance. Exact binomial inversion and bounded-mean concentration are attributed as classical ingredients; the new population conclusion is the specified transfer from actual certified gains. The stopped Brownian verification subcase retains its hypotheses and complete boundary argument. The present two-date validation supplies no proof or experimental substitute for a general coupled NDU/FBSDE, nonlinear recursive-preference application, temporal-self equilibrium or game.

The completed evidence for this item is in Table E.III; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The development retains the original correctness corrigenda and the verified approximation, HJB, recursive-utility and policy-iteration connections at their stated assumptions. The new argument identifies its classical ingredients and the specific original-risk step: constrained-tail marginal regularity, paid outward probe intervals, a finite bounded fitting certificate, and an explicit nonsmooth feature-range transfer. R52 quotient arguments, R67 readout/dual construction and R99 constrained stationarity are credited rather than presented as new here. The original model is not silently replaced by a Brownian utility or a different information filtration.

Remaining scientific requirement. The preserved NDU, FBSDE, time-inconsistent and game claims require their own well-posedness, boundary, equilibrium and execution evidence. The present investment calculation does not certify those extensions.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

2.30 Minor 10. Title and neural content

Original advisory request.

> 10. The title should not carry more neural-specific content than the theorem and evidence identify.

Completed R120 author response. The title and active main text retain Neural Bellman Operators as the study of learned original continuations, recursive operators and feasible actor/critic decisions. The new action-dependent comparison and safeguard are attributed to an eligible-method interface shared by the neural and conventional representations. Their original-policy gains are actually measured and independently bounded; conventional hinge remains identical to ReLU, and the strongest original-formula controls remain visible. This establishes a concrete shared numerical-methods result inside the NBO problem while leaving equal-information neural advantage and action-aware learning as explicit ongoing research requirements.

The completed evidence for this item is in Table E.I; the exact row files and hashes are retained in the current response map.

R121 mathematical and algorithmic action. The title and main claim remain Neural Bellman Operators: learning a recursive continuation and its feasible actor in the original economic decision problem. The current contribution is stated as an original-risk marginal learning construction with a verified finite fitting problem and a complete comparison. The representation-neutral mathematics, the neural implementation and the empirical question of neural value are distinguished. Historical own-future neural operators and strictly qualified stochastic-control/recursive-utility links remain preserved, while unsupported exclusivity and unexecuted adoption are not stated as achieved results.

Remaining scientific requirement. Preserving the research identity is not a claim that every desired neural or economic advantage has already been established. The unresolved scientific requirements continue to guide the revision.

New execution status. The current R121 outcome is pending in the prospective numerical component `new121/numerical121.tex`; the completed R120 evidence above is not a substitute.

3 Historical preservation, unresolved episodes and publication scope

All earlier original report text and author response routes remain unchanged in the compact inherited maps indexed by `provenance121/reports121/REPORT_SOURCE_INDEX121.json`. The 620 major routes, 68 explicit minor requests, 295 separately retained supplementary requirements, 31 embedded requests and 35 overlapping parent treatments total 1,049 authored route records. The overlapping parent treatments are not added as independent scientific issues. Exact source texts and their original commit identities remain authoritative.

R98 and R114/R115 retain their independent protocols, selected-model and sample barriers. A previously recorded not-started state is a dated observation, not a live process-status assertion. R111–R118 retain their separate model and information conventions, incomplete handoffs, failed transport and actual later reproductions where documented. In particular, R119's actual unchanged R117/R118 reproductions do not complete the independent R115 population study. R118's inherited R110 article and truncated transport do not constitute a newer completed author baseline. The retention map records these episode-specific boundaries: `CONTENT_RETENTION_MAP121.md`.

The original-theory and source registries distinguish a complete file or cross-reference census from full semantic reading and from fresh mathematical revalidation. Corrected historical claims remain visible with their corrigenda; their original source is not silently edited. The report reading-coverage receipt records the actual reader and scope of each full report reading rather than inferring reading from a hash.

R120 source CI run 39240515942 actually completed successfully at its bound source and produced artifact 12181820272. Its independently read artifact and exact four-PDF comparison are retained as R120 evidence. They are not R121 CI. The final R121 source, PDF and publication declarations will be separate later records bound to their actual outputs; a queued run, a successful archive copy or a bounded reporting replay will not be described as a complete scientific rerun.